

Notes: Faccio and Lang (JFE, 2002)

The Ultimate Ownership of Western European Corporations

One-sentence summary. This note summarizes Faccio and Lang (2002), which hand-collects and traces ultimate ownership and control chains for 5,232 listed corporations in 13 Western European countries, distinguishing cash-flow rights from voting rights, classifying ultimate owners, and documenting how dual class shares, pyramids, multiple control chains, and cross-holdings shape corporate control; the central finding is that Western European listed firms are mainly either widely held or family controlled, with the United Kingdom and Ireland looking much more dispersed than continental Europe, while economically large discrepancies between ownership and control are concentrated in only a subset of countries.

Contents

1	Big picture	2
1.1	Main question	2
1.2	Core answer	2
1.3	Why this paper matters	2
2	Incremental contribution of the paper	2
2.1	Comprehensive Western European ownership map	2
2.2	Ultimate ownership, not just direct ownership	2
2.3	Separation of cash-flow rights and control rights	3
2.4	Mechanism-level evidence	3
2.5	Country-level heterogeneity	3
3	Data and measurement	3
3.1	Sample	3
3.2	Control thresholds	3
3.3	Owner categories	3
3.4	Key control-enhancing devices	4
4	Main empirical findings	4
4.1	Western Europe is split between family control and dispersed ownership	4
4.2	Financial firms and large firms are more dispersed	4
4.3	Dual class shares are very country-specific	4
4.4	Pyramids are present, but not universally dominant	5
4.5	Family control is often managerial control	5
4.6	Ownership-control wedges are moderate on average but large in selected countries	5
4.7	Family control is concentrated in some countries	5
5	Interpretation of the extracted figures and tables	5
5.1	Table 1: Data sources	5
5.2	Table 2: Sample screening	6

5.3	Figure 1: Nordstrom family group in Sweden	6
5.4	Figure 2: Unicem and the Agnelli group	6
5.5	Table 3: Ultimate control at the 20% threshold	6
5.6	Table 4: Financial versus nonfinancial firms	7
5.7	Table 5: Firm size and control	7
5.8	Table 6: Legal restrictions and dual class shares	7
5.9	Table 7: Control-enhancing devices among controlled firms	7
5.10	Table 8: Control-enhancing mechanisms by owner type	8
5.11	Table 9: Cash-flow rights, control rights, and their ratio	8
5.12	Table 10: Concentration of family control	8
6	Extracted figures and tables from the paper	9
7	Economic interpretation	13
7.1	The main agency problem differs across governance systems	13
7.2	Family control can be stabilizing and problematic at the same time	13
7.3	Country law matters, but law is not enough	13
7.4	Large-firm samples can mislead	13
8	Limitations and caveats	13
8.1	Data are cross-sectional and late-1990s centered	13
8.2	Unlisted firms are often treated as families	13
8.3	A 20% threshold is a convention	14
8.4	The paper maps governance structures rather than outcomes	14
9	Takeaways	14

1 Big picture

1.1 Main question

The paper asks two related questions. First, who ultimately owns and controls listed corporations in Western Europe? Second, through what mechanisms do controlling owners obtain voting control that exceeds their cash-flow ownership? The paper is motivated by the idea that the classic Berle-Means image of widely dispersed ownership may not describe most firms outside the United States, even in rich, developed markets.

1.2 Core answer

The answer is that Western Europe is not uniform. In the full sample, the two dominant categories are **family controlled firms** and **widely held firms**. Family control is especially prevalent in continental Europe. Widely held firms are especially prevalent in the United Kingdom and Ireland. State control matters for some large firms in continental Europe, while control by widely held corporations is rare. Financial firms and large firms are more likely to be widely held; nonfinancial and small firms are much more likely to be family controlled.

1.3 Why this paper matters

The paper changes the unit of analysis from direct block ownership to **ultimate control**. A direct shareholder may be another firm, a bank, the state, or an unlisted company. The economically meaningful question is often not who directly owns shares, but who stands at the end of the control chain. By following chains through corporations and by accounting for voting rights, the paper shows that conventional datasets can misclassify who actually controls firms.

2 Incremental contribution of the paper

2.1 Comprehensive Western European ownership map

Relative to earlier work, this paper provides a much broader and more detailed map of ownership in Western Europe. La Porta et al. focus on a small sample of large firms across many countries; Claessens et al. focus on East Asia. Faccio and Lang extend the ultimate-ownership approach to a large sample of 5,232 listed firms in 13 Western European countries, including many medium and small firms and both financial and nonfinancial firms.

2.2 Ultimate ownership, not just direct ownership

The paper's central methodological contribution is to trace direct blockholders back to their ultimate owners. This is crucial because a listed company can appear to be owned by another corporation, but that corporation may itself be controlled by a family, the state, or a financial institution. The paper's ownership categories are therefore based on control chains rather than simply the names appearing on shareholder lists.

2.3 Separation of cash-flow rights and control rights

The paper measures ownership as cash-flow rights and control as voting rights. If a family owns 25% of Firm X and Firm X owns 20% of Firm Y, the family owns 5% of Firm Y's cash-flow rights but controls 20% of Firm Y through the weakest link in the chain. This distinction allows the paper to identify the extent to which control is achieved with limited cash-flow exposure.

2.4 Mechanism-level evidence

The paper documents the concrete devices that create wedges between ownership and control: dual class shares, pyramids, holdings through multiple control chains, and cross-holdings. This mechanism-level evidence is important because different governance concerns arise from different control devices. A dual class structure creates disproportionate voting power directly; a pyramid creates leverage through chains; multiple control chains aggregate control through several routes; cross-holdings can reinforce control loops.

2.5 Country-level heterogeneity

The paper shows that Western Europe contains multiple governance systems. The United Kingdom and Ireland are relatively dispersed. Continental Europe relies more on controlling owners, especially families. Some countries use dual class shares heavily, while others restrict them. Ownership-control wedges are substantial in Belgium, Italy, Norway, Sweden, and Switzerland, but much smaller elsewhere.

3 Data and measurement

3.1 Sample

The sample begins with 5,547 listed firms in Austria, Belgium, Finland, France, Germany, Ireland, Italy, Norway, Portugal, Spain, Sweden, Switzerland, and the United Kingdom. After excluding firms with missing or unreliable ownership data, foreign-controlled affiliates whose ultimate owner cannot be traced, and firms controlled through nominee accounts, the final sample contains 5,232 firms, or 94.32% of listed firms in the 13 countries.

3.2 Control thresholds

The main classification uses a 20% voting-right threshold. If no shareholder controls at least 20% of votes, the firm is classified as widely held. The paper also discusses a 10% threshold. Lowering the threshold mechanically reduces the fraction of widely held firms and increases the fraction classified as controlled, especially by families and financial institutions.

3.3 Owner categories

Ultimate controlling owners are classified into six categories:

- **Family:** an individual, family, or unlisted firm.
- **Widely held financial institution:** a financial firm that is itself widely held.

- **State:** national government, local authority, or government agency.
- **Widely held corporation:** a widely held nonfinancial corporation.
- **Cross-holdings:** cases in which firms control each other or directly hold their own shares.
- **Miscellaneous:** charities, trusts, employees, cooperatives, or minority foreign owners.

The family category includes unlisted firms whose own owners cannot be traced. The authors justify this convention by examining evidence that unlisted firms in major European countries are overwhelmingly family controlled.

3.4 Key control-enhancing devices

- **Dual class shares:** different classes of shares have different voting rights per unit of cash-flow ownership.
- **Pyramids:** an ultimate owner controls a firm indirectly through another corporation that it does not fully own.
- **Multiple control chains:** an ultimate owner controls a firm through several different paths.
- **Cross-holdings:** the controlled firm participates directly or indirectly in ownership of the controlling firm, or holds a significant block of its own shares.

4 Main empirical findings

4.1 Western Europe is split between family control and dispersed ownership

At the 20% threshold, 44.29% of firms are family controlled and 36.93% are widely held. The United Kingdom and Ireland stand apart: 63.08% of UK firms and 62.32% of Irish firms are widely held. In much of continental Europe, family control is the dominant form. Germany, France, Italy, Portugal, and Spain all show very large family-control shares.

4.2 Financial firms and large firms are more dispersed

Financial firms are more likely to be widely held than nonfinancial firms. They are also much more likely to be controlled by widely held financial institutions. By size, large firms are more likely than small firms to be widely held, while small firms are more likely to be family controlled. This size pattern is important because large-firm samples alone can overstate the importance of dispersed ownership and understate the importance of family control.

4.3 Dual class shares are very country-specific

Dual class shares are rare in some countries but very common in others. Sweden, Switzerland, Italy, Finland, Ireland, Austria, and the United Kingdom show meaningful use of dual class or limited-voting structures. Legal restrictions matter, but they are not the whole story: even among countries with formal restrictions, practice differs substantially.

4.4 Pyramids are present, but not universally dominant

Among firms with a controlling shareholder at the 20% level, 19.13% use pyramids, 5.52% use multiple control chains, and only 0.73% use cross-holdings. This means that the stereotypical European control structure is not always a pyramid. The more basic fact is the presence of a controlling owner; the control-enhancing device varies by country and owner type.

4.5 Family control is often managerial control

In 68.45% of family-controlled firms, a member of the controlling family is in top management. This is a crucial governance point. Family control is not only ownership control; it frequently means direct managerial involvement. The agency problem is therefore not only managers versus dispersed shareholders, but often controlling families versus minority shareholders.

4.6 Ownership-control wedges are moderate on average but large in selected countries

For firms with at least one shareholder holding at least 5% of voting rights, the largest ultimate owner holds, on average, 34.64% of cash-flow rights and 38.48% of voting rights. The average ratio of cash-flow to voting rights is 0.868. The wedge is largest in Switzerland, Italy, Norway, Sweden, and Belgium, and much smaller in Spain, Portugal, and France.

4.7 Family control is concentrated in some countries

The average number of listed firms controlled per family is not extremely high in most countries, but the market value controlled by top families can be large. The top family controls 17.89% of listed market value in Switzerland and 10.40% in Italy. The top 15 families control over one-third of market value in Portugal and Belgium, but far less in the United Kingdom and Spain.

5 Interpretation of the extracted figures and tables

5.1 Table 1: Data sources

Table 1 documents how the ownership data were assembled. This is not just a technical appendix. The key point is that ownership disclosure systems differ by country, so a single commercial database would not be reliable enough for the paper’s purpose. The authors combine stock-exchange files, official regulatory records, annual-report databases, newspapers, Worldscope, Datastream, and country-specific ownership handbooks.

The interpretation is that the paper’s dataset is a major contribution in itself. The table also warns the reader that ultimate ownership studies are only as credible as the tracing procedure. Countries such as Spain and Italy benefit from official disclosure sources; other countries require cross-checking across multiple sources. The table explains why several countries are excluded and why the sample focuses on countries where ownership can be reliably reconstructed.

5.2 Table 2: Sample screening

Table 2 shows that the authors start with 5,547 listed firms and retain 5,232 after screening, or 94.32% of the initial universe. The exclusions are small: 3.01% have no ownership data, 1.57% are majority controlled by foreign investors whose ultimate chain cannot be traced, and 1.10% are controlled by nominees.

The table also previews the samples used later. All 5,232 firms are used for the main ownership tables; 4,806 firms have at least one 5% shareholder and are used for cash-flow/control-rights analysis; 3,300 firms have at least one 20% shareholder and are used for control-enhancement tables. The interpretation is that the paper is not relying on a narrow selected sample. It covers almost the entire listed universe of the included countries.

5.3 Figure 1: Nordstrom family group in Sweden

Figure 1 is a worked example showing how dual class shares and pyramiding operate. Panel A maps the Nordstrom family group and the ownership/control stakes along the links. Panel B shows how the Realia share structure transforms cash-flow ownership into voting control. A-shares represent only 5.8% of capital but 38.09% of votes because each A-share has ten times the voting power of each B-share.

The economic lesson is that a small cash-flow position can translate into much greater control when high-vote shares and control chains interact. Columna owns 35.0% of Realia's cash-flow rights but controls 48.1% of votes. Through Columna, the Nordstrom family controls Realia even though its ultimate cash-flow stake is much smaller than its control stake. The figure explains why direct ownership percentages are insufficient for governance analysis.

5.4 Figure 2: Unicem and the Agnelli group

Figure 2 provides a more complex Italian example. The Agnelli family controls Unicem through a combination of pyramiding, dual class shares, and multiple control chains. The family ultimately owns only 9.04% of Unicem's cash-flow rights but controls 47.64% of the votes.

This is one of the most important illustrations in the paper because it shows how several mechanisms compound. The hard lines represent pyramidal links, while dotted lines represent additional control chains. The control stake is calculated using weakest links along each chain and summing across valid chains. The result is a very large wedge between economic ownership and voting control. This is the kind of structure that raises minority-shareholder concerns because the controlling family has strong control but limited cash-flow exposure.

5.5 Table 3: Ultimate control at the 20% threshold

Table 3 is the central descriptive table. It shows that family control and widely held ownership dominate Western Europe. The total row reports 36.93% widely held and 44.29% family controlled. The country rows reveal the Anglo-continental split: the United Kingdom and Ireland are mostly widely held, while most continental countries have majority or near-majority family control.

The table also shows that state control is concentrated in particular countries and is not a pan-European default. State control exceeds 10% in Austria, Finland, Italy, and Norway. Widely held financial institutions are meaningful in Belgium, France, Italy, Spain, Switzerland, and the

United Kingdom, but control by widely held nonfinancial corporations is small almost everywhere. The interpretation is that Europe's governance problem is usually about controlling shareholders, especially families, not managerial control of widely held firms.

5.6 Table 4: Financial versus nonfinancial firms

Table 4 shows that ownership structure depends on industry type. Financial firms are more likely to be widely held and much more likely to be controlled by widely held financial institutions. In the full sample, 39.92% of financial firms are widely held versus 36.39% of nonfinancial firms; 21.48% of financial firms are controlled by widely held financial institutions versus only 5.96% of nonfinancial firms.

The interpretation is that financial-sector regulation, business models, and institutional history make financial firms different. This matters because countries with more financial listings can appear more dispersed even if their nonfinancial corporate sector remains family dominated. In other words, cross-country ownership comparisons are affected by the sectoral composition of listed firms.

5.7 Table 5: Firm size and control

Table 5 shows that size strongly predicts ownership structure. The largest firms are more likely to be widely held, while the smallest firms are more likely to be family controlled. In the total rows, 45.24% of large firms are widely held, compared with 29.28% of middle firms and 25.32% of small firms. Family control rises from 27.24% among the largest firms to 59.66% among the smallest firms.

This table is important for interpreting earlier international evidence. Studies based only on the largest firms can overstate how common dispersed ownership is. The table also shows where the state matters: among the largest firms in Austria, Finland, Italy, Norway, and Portugal, state ownership is economically meaningful. The broad lesson is that ownership systems look more dispersed at the top of the size distribution and more family dominated among the broader listed universe.

5.8 Table 6: Legal restrictions and dual class shares

Table 6 links legal rules to the use of dual class shares. The variable $\text{Own} = 20\% \text{ Con}$ measures the minimum percentage of book equity needed to control 20% of votes. A lower value means that voting control can be achieved with less cash-flow ownership. Dual class shares are especially prevalent in Sweden, Switzerland, Italy, Finland, Ireland, Austria, and the United Kingdom.

Panel B shows that countries requiring a minimum voting ratio or allowing unrestricted dual class structures have much more use of dual class shares than countries that cap or prohibit nonvoting shares. The interpretation is that law matters, but legal categories do not perfectly determine outcomes. Market practice, historical ownership norms, and country-specific institutions also shape how aggressively firms separate votes from capital.

5.9 Table 7: Control-enhancing devices among controlled firms

Table 7 focuses only on the 3,300 firms with a controlling shareholder at the 20% threshold. Among these firms, 19.13% use pyramids, 5.52% use multiple control chains, and 0.73% use cross-holdings. Control is often held by a single controlling owner: 53.99% of controlled firms have no other owner with at least 10% of votes.

The table also reports that 68.45% of family-controlled firms have a family member in top management. The country variation is substantial: pyramids are very high in Norway, Belgium, Germany, and the United Kingdom; multiple chains are particularly important in Norway, Italy, and Germany. The key interpretation is that family-controlled firms often combine ownership, voting control, and management control, while the formal mechanisms used to enhance control vary strongly by country.

5.10 Table 8: Control-enhancing mechanisms by owner type

Table 8 separates family-controlled, state-controlled, and widely held financial institution-controlled firms. Family-controlled firms use pyramids in 13.81% of cases and multiple control chains in 3.22%. State-controlled firms use pyramids in 35.32% of cases and multiple control chains in 11.01%. Widely held financial institution-controlled firms use pyramids in 27.96% and multiple chains in 16.78%.

The interpretation is that families are not the only users of complex control structures. The state and financial institutions use pyramids and multiple chains more often than families. This is important because ownership-control wedges are not only a family-capitalism issue. They can also arise in state and bank-centered systems. The table also shows that family control is often managerial, while state and financial-institution control are more organizational forms of control.

5.11 Table 9: Cash-flow rights, control rights, and their ratio

Table 9 measures the wedge between ownership and control. The largest ultimate shareholder holds an average of 34.64% of cash-flow rights and 38.48% of voting rights. The total average ratio of cash-flow to control rights is 0.868. A ratio below one means voting control exceeds economic ownership.

Country differences are the key result. The ratio is lowest in Switzerland (0.740), Italy (0.743), Norway (0.776), Belgium (0.779), and Sweden (0.790), indicating more substantial separation between ownership and control. The ratio is highest in Spain (0.941), France (0.930), and Portugal (0.924), indicating smaller wedges. The table therefore identifies where the agency problem between controlling and minority shareholders is likely to be most severe.

5.12 Table 10: Concentration of family control

Table 10 studies whether a few families dominate national stock markets. The average number of listed firms controlled by each family is modest in most countries, but Italy stands out with 1.46 firms per family. The market-value concentration of family control is more striking: the top family controls 17.89% of listed market value in Switzerland and 10.40% in Italy.

The top 15 families control 36.77% of listed market value in Portugal and 36.63% in Belgium, but only 6.55% in the United Kingdom, 13.48% in Spain, and 15.38% in Ireland. The interpretation is that family capitalism can matter even when each family controls only a small number of listed firms, because the firms they control may be very large. The table also shows why country-level governance debates often focus on specific business groups or families.

6 Extracted figures and tables from the paper

Table 1
List of country specific data sources used to collect data on direct ownership (i.e., names of large shareholders, and stakes held in the company), and class share structures.

Countries	Direct ownership data	Dual class shares
Austria	Wiener Börse (2000), "Yearbook 1999"	Wiener Börse (2000), "Yearbook 1999"
Belgium	Brussels Stock Exchange (www.bss.be) Worldscope (1998)	Datastream Brussels Stock Exchange
Finland	Helsinki Media Blue Book, "Major Finnish Firms Internet Database" (http://www.bluebook.fi/en/nexttest/doku.php?id=main:finnish_firms) Hugin, Annual Report CD (1998) (http://www.huginonline.com)	Datastream Helsinki Media Blue Book, "Major Finnish Firms I Database"
France	The Herald Tribune (1997), "French Firm Handbook 1997," SFB-Paris Bourse Financial Times (1997), "Extel Financial"	Datastream Financial Times (1997), "Extel Financial" Les Echos (1996) Musa (1998)
Germany	http://www.bourse-de-paris.fr/fr/market8/fsg30.htm Commerzbank (1997), "Wer gehört zu wem?" (http://www.commerzbank.com/navigate/date_frm.htm) Financial Times (1997), "Extel Financial"	Datastream Die Welt (1996), available at: http://www.welt.de Financial Times (1997), "Extel Financial"
Ireland	Worldscope (1998) CONSOB (1997), "Bollettino - edizione speciale n. 4/97 - Compagine azionaria delle società quotate in borsa o ammesse alle negoziazioni nel mercato ristretto al 31 dicembre 1996" (http://www.consob.it/trasparenza_soc_quot/trasp_soc_quot.htm) Il Sole 24 ore (1997), "Il taccuino dell'azionista"	Datastream Il Sole 24 ore (1997), "Il taccuino dell'azionista"

Table 1, part 1: country-specific data sources for direct ownership and dual class shares.

Portugal	Bolsa de Valores de Lisboa, "Sociedades Cotadas 1997" (1998)	Hugin, Annual Report CD (1998) Bolsa de Valores de Lisboa, "Sociedades Cotadas 1997"
Spain	Comisión Nacional del Mercado de Valores (1998), "Participaciones significativas en sociedades cotizadas" (http://www.cnmv.es/english/cnmv.htm)	ABC (1996) Financial Times (1997), "Extel Financial"
Sweden	Hugin, Annual Report CD (1998) Worldscope (1998)	Crespi-Cladera and García-Castón (1998) Datastream Hugin, Annual Report CD (1998)
Switzerland	Union Bank of Switzerland (1998), "Swiss Stock Guide 96/97," Zürich	Union Bank of Switzerland (1998), "Swiss Stock Guide 96/97," Zürich Datastream
United Kingdom	Financial Times (1997), "Extel Financial" London Stock Exchange (1997), "The London Stock Exchange Yearbook"	Financial Times (1997), "Extel Financial" Financial Times (1996)

Table 1, part 2: continuation of data sources.

Table 2
Screenings used to identify and remove corporations without reliable ownership data, and sub-samples of firms used in the different sections of the paper. Panel A describes the procedure used to eliminate companies with insufficient or unreliable data on ownership, due to unavailable data, or related to the impossibility to trace ownership back to the ultimate owner (i.e., when the direct controlling shareholder is from a country for which we do not have ownership data, or when the direct owner is a nominee). Panel B reports the total number of firms remaining after this screening, as well as the number of firms with an ultimate owner at the 5% and 20% cutoffs. It also lists the different subsets of firms used in the various sections of the paper.

Panel A: Selection criteria	Number	Percentage
All listed firms	5,547	100.00
Firms with no ownership data	(167)	(3.01)
Firms majority-controlled by foreign investors	(87)	(1.57)
Firms controlled by "nominees"	(61)	(1.10)
Firms in our sample	5,232	94.32

Panel B: Sample of firms analyzed in the various tables	Number	Percentage	Tables
All listed firms	5,232	100.00	3, 4, 5, 6
Firms where at least one shareholder holds at least 5% of voting rights	4,806	91.86	9
Firms where at least one shareholder holds at least 20% of voting rights	3,300	63.07	7, 8
Selected samples of firms			10; Figs. 1 and 2

Table 2: sample screening and subsamples used in later tests.

M. Faccio, L. L. F. Lang / Journal of Financial Economics 62 (2002) 297–332

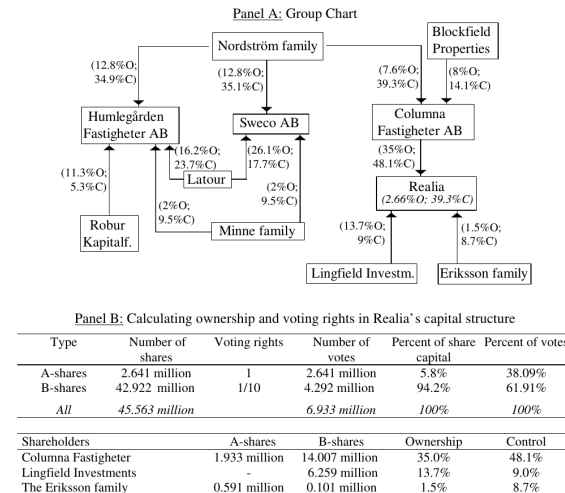


Fig. 1. The Nordström family group (Sweden). This figure describes the major listed firms controlled by the Nordström family. All control stakes of at least 5% are reported. All the firms have dual class shares with A-shares each carrying one voting right and B-shares each carrying a tenth of a vote. The procedure used to compute direct ownership and control rights is illustrated in Panel B. Ownership stakes are denoted by "O" and voting stakes by "C." Direct ownership and control stakes are shown alongside each of the arrows, while the resulting ultimate ownership and control stakes for Realia (the only case of pyramiding) by the Nordström family are reported in the firm's box.

Figure 1: Nordstrom family group and Realia voting-right calculation.

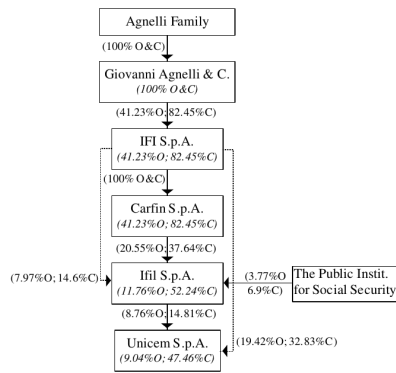


Fig. 2. Unicem (Italy). This figure describes the principal shareholders of Unicem, a complex case of pyramiding, holdings through multiple control chains, and the use of dual class shares within the Agnelli family group, the largest Italian group. All control stakes of at least 5% are reported. The company has only one ultimate owner, the Agnelli family, which controls 47.64% of the voting rights and owns 9.04% of the cash-flow rights in Unicem. Hard lines indicate pyramiding while dotted lines indicate holdings through further (i.e., multiple) control chains. Ownership stakes are denoted by “O” and voting stakes by “C.” Direct ownership and control stakes are shown alongside each arrow, while the Agnelli family’s

Figure 2: Unicem control chain in the Agnelli group.

Table 4

Concentration of control: financial versus non financial firms. Data relating to 5,232 publicly traded corporations are used to construct this table groups corporations into financial firms (SIC: 6000-6999) and non financial corporations. The table presents the percentage of firms controlled by c controlling owners at the 20% threshold. Controlling shareholders are classified into six types: Family: A family (including an individual) or a firm unlisted on any stock exchange. Widely held financial institution: A financial firm (SIC 6000-6999) that is widely held at the control threshold. State: A national government (domestic or foreign), local authority (county, municipality, etc.), or government agency. Widely held corporation: A non financial widely held at the control threshold. Cross-holdings: The firm Y is controlled by another firm, that is controlled by Y, or directly controls at least 20% own stocks. Miscellaneous: Charities, voting trusts, employees, cooperatives, or minority foreign investors. Companies that do not have a share controlling at least 20% of votes are classified as widely held.

Country	Type of firm	% of firms	Widely held	Family	State	Widely held corporation	Widely held financial	Miscellaneous	Cross-h
Austria	Financial	30.3	10.00	38.33	13.33	0.00	18.33	16.67	3.
	Non-financial	69.7	11.59	59.18	16.18	0.00	4.55	8.70	0.
Belgium	Financial	39.2	37.25	32.35	1.96	1.96	13.73	12.75	0.
	Non-financial	60.8	8.86	63.92	2.53	0.00	12.03	12.66	0.
Finland	Financial	10.1	23.08	64.10	10.26	0.00	2.56	0.00	0.
	Non-financial	89.9	31.71	44.11	17.07	1.22	0.61	5.28	0.
France	Financial	16.1	22.45	33.16	6.63	0.00	36.22	1.53	0.
	Non-financial	83.9	12.38	70.92	4.81	4.52	6.58	0.79	0.
Germany	Financial	14.5	12.75	37.58	5.88	0.00	28.27	1.96	13.
	Non-financial	85.5	9.07	69.20	6.37	4.26	5.82	3.61	0.
Ireland	Financial	15.9	63.64	13.64	9.09	4.55	0.00	9.09	0.
	Non-financial	84.1	62.07	26.72	0.00	1.72	5.17	4.31	0.
Italy	Financial	23.6	24.49	18.37	8.16	7.14	38.78	0.00	3.
	Non-financial	76.4	9.43	72.33	11.01	1.57	4.09	1.57	0.
Norway	Financial	16.1	56.00	12.00	24.00	0.00	4.00	4.00	0.
	Non-financial	83.9	33.08	43.65	10.99	0.38	4.55	4.64	2.
Portugal	Financial	24.1	19.05	71.43	0.00	0.00	4.76	4.76	0.
	Non-financial	75.9	23.44	55.47	7.51	0.78	4.69	7.51	0.
Spain	Financial	16.1	29.41	23.37	7.35	0.00	38.40	1.47	0.
	Non-financial	83.9	25.85	62.03	3.49	1.95	6.34	0.28	0.
Sweden	Financial	17.1	50.00	40.48	4.76	0.00	2.38	2.38	0.
	Non-financial	82.9	37.43	47.49	5.03	0.00	3.35	6.70	0.
Switzerland	Financial	24.3	32.69	24.04	17.31	0.00	16.35	9.62	0.
	Non-financial	75.7	25.93	55.86	4.12	1.44	7.10	5.25	0.
UK	Financial	0.185	60.11	18.01	0.00	0.42	17.94	3.53	0.
	Non-financial	0.815	63.76	24.97	0.09	0.84	6.90	3.44	0.
All	Financial	0.183	39.92	26.54	5.05	1.31	21.48	4.00	1.
	Non-financial	0.817	36.39	48.15	3.85	2.13	5.96	3.30	0.

Table 4: control concentration by financial versus nonfinancial firms.

Table 3

Ultimate control of publicly traded firms. Data relating to 5,232 publicly traded corporations are used to construct this table. The table presents the per of firms controlled by different controlling owners at the 20% threshold. Controlling shareholders are classified into six types: Family: A family (incl individual) or a firm that is unlisted on any stock exchange. Widely held financial institution: A financial firm (SIC 6000-6999) that is widely held at the threshold. State: A national government (domestic or foreign), local authority (county, municipality, etc.), or government agency. Widely held corporation: A non financial firm, widely held at the control threshold. Cross-holdings: The firm Y is controlled by another firm, that is controlled by Y, or directly controls at least 20% of its own stocks. Miscellaneous: Charities, voting trusts, employees, cooperatives, or minority foreign investors. Companies that do not shareholder controlling at least 20% of votes are classified as widely held.

Country	Number of Firms	Widely held	Family	Family of which:		State	Widely held Corporation	Widely held Financial	Miscellaneous	Cross-h
				Identified families	Unlisted firms					
Austria	99	11.11	52.86	12.12	40.74	15.32	0.00	8.59	11.11	1.
Belgium	130	20.00	51.54	7.31	44.23	2.31	0.77	12.69	0.	0.
Finland	129	28.68	48.84	16.20	32.56	15.76	1.55	0.65	4.52	0.
France	607	14.00	64.82	26.11	38.71	5.11	3.79	11.37	0.91	0.
Germany	704	10.37	64.62	27.03	37.59	6.30	3.65	9.07	3.37	2.
Ireland	69	62.32	24.63	13.04	11.59	1.45	2.17	4.35	5.07	0.
Italy	208	12.98	59.61	39.50	20.11	10.34	2.88	12.26	1.20	0.
Norway	155	36.77	38.55	10.59	27.96	13.09	0.32	4.46	4.54	2.
Portugal	87	21.84	60.34	5.17	55.17	5.75	0.57	4.60	6.90	0.
Spain	632	28.42	55.79	6.25	49.54	4.11	1.64	11.51	0.47	0.
Sweden	245	39.18	46.94	22.65	24.29	4.90	0.00	2.86	5.71	0.
Switzerland	214	27.57	48.13	22.66	25.47	7.32	1.09	9.35	6.31	0.
UK	1,953	63.08	23.68	12.22	11.46	0.08	0.76	8.94	3.46	0.
Total	5,232	36.93	44.29	16.93	27.36	4.14	1.68	9.03	3.43	0.

Table 3: ultimate control of publicly traded firms at the 20% threshold.

Table 5

Concentration of control and firm size. Data relating to 5,232 publicly traded corporations are used to construct this table, which groups corps according to their size (market capitalization), and, for each variable, presents means for the largest 20 firms, 50 medium-size firms, and the smallest 5. For countries with less than 120 firms, we group the largest 20 firms in the “Largest 20” group, and then divide the rest into two equal groups. T presents the percentage of firms controlled by different controlling owners at the 20% threshold. Controlling shareholders are classified into six types. A family (including an individual) or a firm that is unlisted on any stock exchange. Widely held financial institution: A financial firm (SIC 6000-6999) widely held at the control threshold. State: A national government (domestic or foreign), local authority (county, municipality, etc.), or government Widely held corporation: A non financial firm, widely held at the control threshold. Cross-holdings: The firm Y is controlled by another firm, controlled by Y, or directly controls at least 20% of its own stocks. Miscellaneous: Charities, voting trusts, employees, cooperatives, or minority investors. Companies that do not have a shareholder controlling at least 20% of votes are classified as widely held.

Country	Category	Average market cap (in US\$)	Widely held	Family	State	Widely held corporation	Widely held financial	Misc.	Cross-h
Austria	Largest 20	1,593.0	10.00	32.50	42.50	0.00	5.00	10.00	0.
	Middle 40	227.8	14.29	47.14	11.43	0.00	15.71	8.57	2.
	Smallest 39	33.5	11.76	71.57	4.90	0.00	2.94	8.82	0.
Belgium	Largest 20	5,386.4	20.00	37.50	0.00	0.00	27.50	15.00	0.
	Middle 50	722.2	18.00	47.00	4.00	0.00	20.00	11.00	0.
	Smallest 50	58.0	16.00	65.00	0.00	2.00	2.00	15.00	0.
Finland	Largest 20	2,310.2	38.10	33.33	28.57	0.00	0.00	0.00	0.
	Middle 50	274.3	41.18	38.23	11.76	2.94	1.47	4.41	0.
	Smallest 50	30.1	17.65	65.20	9.80	0.00	0.00	7.35	0.
France	Largest 20	16,467.1	60.00	30.00	0.00	0.00	10.00	0.00	0.
	Middle 50	149.9	14.00	68.00	6.00	2.00	10.00	0.00	0.
	Smallest 50	6.8	8.00	77.00	2.00	4.00	9.00	0.00	0.
Germany	Largest 20	27,510.2	45.00	15.00	10.00	0.00	12.50	5.00	12.
	Middle 50	158.3	10.00	75.00	2.00	7.00	4.00	1.00	1.
	Smallest 50	12.0	14.00	81.00	3.00	0.00	2.00	0.00	0.
Ireland	Largest 20	2,260.2	70.00	20.00	0.00	2.50	0.00	7.50	0.
	Middle 25	198.9	50.00	29.16	0.00	4.17	8.33	0.00	0.
	Smallest 24	37.0	68.00	24.00	4.00	0.00	4.00	0.00	0.
Italy	Largest 20	10,398.8	35.00	20.00	25.00	0.00	15.00	0.00	5.
	Middle 50	320.4	8.00	63.00	6.00	6.00	14.00	2.00	1.
	Smallest 50	70.4	14.00	67.00	6.00	2.00	10.00	1.00	0.

Table 5, part 1: control concentration by firm size.

Norway	Largest 20	2,302.7	30.00	45.00	20.00	0.00	0.00	5.00	0.
	Middle 50	310.8	38.00	34.83	16.50	0.00	3.83	5.67	1.
	Smallest 50	62.4	40.00	35.67	7.40	0.00	7.33	4.40	5.
Portugal	Largest 20	2,338.9	15.00	60.00	17.50	2.50	0.00	5.00	0.
	Middle 34	170.4	28.57	60.00	2.86	0.00	5.71	2.86	0.
	Smallest 33	42.8	18.75	60.94	1.56	0.00	6.25	12.50	0.
Spain	Largest 20	4,919.2	45.00	20.83	10.00	9.17	15.00	0.00	0.
	Middle 50	302.4	34.00	46.00	8.00	2.00	10.00	0.00	0.
	Smallest 50	65.3	36.00	56.00	2.00	0.00	6.00	0.00	0.
Sweden	Largest 20	9,575.2	80.00	5.00	5.00	0.00	0.00	10.00	0.
	Middle 50	397.9	49.02	42.16	2.94	0.00	0.98	4.90	0.
	Smallest 50	44.0	32.00	63.00	0.00	0.00	3.00	2.00	0.
Switzerland	Largest 20	24,589.9	50.00	35.00	0.00	0.00	15.00	0.00	0.
	Middle 50	352.2	18.00	58.00	10.00	2.00	5.00	7.00	0.
	Smallest 50	58.0	28.00	55.00	7.33	2.67	6.00	1.00	0.
UK	Largest 20	40,563.4	90.00	0.00	0.00	0.00	10.00	0.00	0.
	Middle 50	151.7	64.71	19.60	0.00	0.00	13.73	1.96	0.
	Smallest 50	4.2	42.00	38.67	0.00	0.00	8.00	11.33	0.
All	Largest 20	12,027.2	45.24	27.24	12.20	1.09	8.46	4.42	1.
	Middle 50	295.0	29.28	48.82	6.53	2.01	8.65	4.25	0.
	Smallest 50	38.9	25.32	59.66	3.72	0.90	5.17	4.80	0.

Table 5, part 2: continuation of size patterns.

Italy	Proportion of non-voting stocks capped	Non voting (and limited voting) capital may not exceed 50% of stock capital	208	18.38	41.
Finland	Minimum vote ratio	Minimum vote ratio: one-tenth	129	15.42	37.
Switzerland	Unrestricted		214	15.26	51.
Sweden	Minimum vote ratio	Minimum vote ratio: one-tenth	245	9.83	66.
All			5,232	18.74	19.

Panel B: Results by type of restrictions

Restrictions	Oen = 20% Con	Dual class shares (%)
Minimum voting ratio required	12.26 ^{U, C, P}	53.92 ^{U, C, P}
Unrestricted	16.86 ^{U, C, P}	40.11 ^{U, C, P}
Cap on the proportion of non voting stocks	19.47 ^{U, M}	10.14 ^{U, M, P}
Prohibited	19.56 ^{U, M}	6.15 ^{U, M, C}

U, M, C, P indicates that the ratio is significantly different (at the 5% level) from the sample of firms in countries where regulation is unrestricted (U), no minimum voting ratio (M), imposes a cap (C) to the proportion of non-voting stocks (with reference to the total stock capital), or prohibits (P) the issue of non voting/limited-voting stocks. The UK is not included in any of these groups since its regulations prohibit the issue of stocks that carry no voting but permit the issue of limited voting stocks (i.e., preference shares). Information about restrictions on voting rights is obtained from the nation's exchanges (or their supervisory authorities) as well as from the managers discussed in Section 4.

Table 6, part 2: continuation and results by type of restriction.

Table 6
Legal restrictions on issuing dual class shares. Data relating to 5,232 publicly traded corporations are used to construct this table. *Oen* = 20% *Con* average minimum percent of the book value of equity required to control 20% of votes (as computed for each firm in our sample); *Dual class shares* (%) average percentage of firms with outstanding dual class shares.

Panel A: Country-level results						
Country	Restrictions	Restriction details	Number of firms	Oen = 20% Con	Dual share	
Belgium	One-share-one-vote	Non voting (and limited voting) capital may not exceed 50% of stock capital	130	20.00	0.0	
Portugal	Proportion of non-voting stocks capped	Non voting (and limited voting) capital may not exceed 50% of stock capital	87	20.00	0.0	
Spain	Proportion of non voting stocks capped	Non voting (and limited voting) capital may not exceed 25% of stock capital	632	20.00	0.	
France	Proportion of non voting stocks capped	Minimum voting rights required for preference stocks include voting (1) when their dividend is in arrears, (2) on resolutions for reducing share capital and winding-up the firm, and (3) on resolutions which are likely to affect their class rights	607	19.93	2.4	
UK	Non voting shares outlawed since 1968. Firms may issue "preference shares" provided these are given "adequate voting rights"	Governmental approval needed for exceptions	1,953	19.14	23.	
Norway	One-share-one-vote rule		155	19.05	13.	
Austria	Unrestricted		99	18.96	23.	
Ireland	Unrestricted		69	18.91	28.	
Germany	Proportion of non-voting stocks capped	Non voting (and limited voting) capital may not exceed 50% of stock capital	704	18.83	17.	

Table 6, part 1: legal restrictions and dual class share use.

Table 7
Percentage of firms adopting control-enhancing devices. We only include the 3,300 firms with controlling shareholders at the 20% level. *Pyramids* reports the percentage of firms whose largest controlling shareholder adopts pyramids as control devices. *Holdings through multiple control chains* reports the percentage of firms whose largest controlling shareholder adopts at least 5% of holdings through multiple control chains as control devices. *Cross-holdings* reports the percentage of firms whose largest controlling shareholder adopts cross-holdings as control devices. *Controlling owner alone* reports the percentage of firms that have single controlling owners. *Management* reports the percentage of firms whose top managers come from the largest shareholder's family. The table presents the percentage of firms controlled by different controlling owners at the 20% threshold.

Country	Number of firms	Pyramids	Holdings through multiple control chains	Cross-holdings	Controlling owner alone	Management
Austria	88	20.78	6.49	1.14	81.82	80.00
Belgium	104	25.00	2.38	0.00	71.15	80.00
Finland	92	7.46	1.49	0.00	41.30	69.23
France	522	15.67	2.87	0.00	64.75	62.20
Germany	631	22.89	7.22	2.69	59.90	61.46
Ireland	26	9.09	0.00	0.00	42.31	77.78
Italy	181	20.27	8.78	1.13	58.76	70.00
Norway	98	33.90	20.34	2.04	38.78	66.67
Portugal	68	10.91	0.00	0.00	60.29	50.00
Spain	465	16.00	5.43	0.22	44.30	62.50
Sweden	149	15.91	0.00	0.67	48.32	73.47
Switzerland	155	10.91	0.91	0.00	68.39	70.00
UK	721	21.13	4.93	0.00	43.00	75.85
Total	3,300	19.13	5.52	0.73	53.99	68.45

Table 7: pyramids, multiple control chains, cross-holdings, single controllers, and family management.

Table 8

Means of enhancing control in Europe by types of controlling owners. We only include 3,012 firms with controlling shareholders at the 20% (2,332 firms controlled by families, 218 firms controlled by the State, and 462 firms controlled by widely held financial institutions) level. *Own = 20%*: *Con* reports the average minimum percent of the book value of equity required to control 20% of votes. *Dual class shares (%)* reports the proportion of firms with dual class shares outstanding. *Pyramids* reports the percentage of firms whose largest controlling shareholder adopts pyramids as control devices. *Holdings through multiple control chains* reports the percentage of firms whose largest controlling shareholder adopts at least 5% of holdings through multiple control chains as control devices. *Controlling owner alone* reports the percentage of firms that have single controlling owners. Panel A presents percentages for family controlled firms, Panel B presents percentages for State-controlled firms, and Panel C presents percentages for firms controlled by widely held financial firms.

Country	Number of firms	<i>Own = 20% Con</i>	Dual class shares (%)	Pyramids	Holdings through multiple control chains	Controlling owner alone
<i>Panel A: Family-controlled firms</i>						
Austria	51	18.91	21.57	11.76	1.96	86.27
Belgium	67	20.00	0.00	8.96	4.48	61.19
Finland	65	14.82	43.55	6.15	1.54	36.92
France	395	19.96	1.01	13.16	3.04	63.80
Germany	460	18.75	18.26	14.57	3.04	66.96
Ireland	17	19.53	16.67	5.88	0.00	35.29
Italy	122	18.34	42.62	20.34	5.93	54.55
Norway	60	18.33	21.28	23.33	5.00	38.33
Portugal	54	20.00	0.00	16.67	3.70	61.11
Spain	351	20.00	0.00	11.97	5.41	39.32
Sweden	117	9.34	67.12	18.80	0.85	47.01
Switzerland	106	13.52	68.57	7.55	1.89	65.09
UK	467	19.16	18.84	13.70	2.14	46.47
Total	2,332	18.64	17.61	13.81	3.22	54.74
<i>Panel B: State-controlled firms</i>						
Austria	16	19.67	12.50	37.50	18.75	68.75
Belgium	3	20.00	0.00	33.33	0.00	100.00
Finland	19	16.81	21.05	5.26	0.00	47.37
France	30	19.93	3.33	36.67	3.33	73.33
Germany	41	19.90	4.88	43.90	4.88	17.07
Ireland	1	20.00	0.00	0.00	0.00	100.00
Italy	22	18.50	40.91	27.27	9.09	77.27
Norway	24	20.00	0.00	70.83	58.33	37.50
Portugal	5	20.00	0.00	0.00	0.00	80.00
Spain	26	20.00	0.00	57.69	3.85	88.46
Sweden	12	10.92	62.50	0.00	0.00	75.00
Switzerland	16	18.09	25.00	6.25	0.00	81.25
UK	3	20.00	0.00	33.33	33.33	0.00

Table 8, part 1: control-enhancing devices by family and state owners.

Table 9

Cash-flow and control rights. This table includes data relating to 4,806 publicly traded corporations (including both financial institutions and non financial institutions) where the largest controlling owner has at least 5% of voting rights. *Cash-flow rights* represents the ultimate ownership stake held by the largest controlling shareholder. *Control rights* the percentage of voting rights controlled by the largest controlling shareholder.

Country	Number of firms	Mean	Standard deviation	Median	1st quartile	3rd quartile
<i>Panel A: Cash-flow rights</i>						
Austria	95	47.16	23.52	50.00	25.50	65.00
Belgium	120	35.14	24.96	36.10	14.98	51.81
Finland	119	32.98	23.94	27.60	14.60	49.91
France	604	46.68	26.69	48.98	24.69	66.00
Germany	690	48.54	31.46	48.89	21.05	75.00
Ireland	68	18.82	17.32	14.24	6.76	26.03
Italy	204	38.33	25.13	39.68	16.61	56.83
Norway	149	24.39	21.26	19.42	8.91	36.12
Portugal	86	38.42	20.45	39.31	19.83	52.00
Spain	610	42.72	30.46	32.55	18.50	64.91
Sweden	244	25.15	23.06	17.30	9.45	33.55
Switzerland	189	34.66	24.69	29.00	12.85	51.00
UK	1,628	22.94	17.87	16.21	10.96	29.66
Total	4,806	34.64	26.76	25.90	13.02	51.00
<i>Panel B: Control rights</i>						
Austria	95	53.52	22.77	54.70	34.00	75.00
Belgium	120	40.09	23.20	39.56	19.49	55.86
Finland	119	37.43	22.44	33.70	20.80	52.36
France	604	48.32	25.55	50.00	28.70	66.00
Germany	690	54.50	28.70	50.76	27.00	76.91
Ireland	68	21.55	16.39	16.64	10.39	26.56
Italy	204	48.26	21.00	50.11	31.39	63.15
Norway	149	31.47	20.18	27.78	15.10	43.59
Portugal	86	41.00	19.18	44.95	22.28	52.30
Spain	610	44.24	29.59	35.73	20.00	65.03
Sweden	244	30.96	22.37	24.90	14.50	40.55
Switzerland	189	46.68	25.97	50.00	22.50	63.00
UK	1,628	25.13	17.87	18.02	13.28	30.19
Total	4,806	38.48	26.10	30.01	15.96	53.98
<i>Panel C: Ratio of cash flow to control rights</i>						
Austria	95	0.851	0.224	1.000	0.704	1.000
Belgium	120	0.779	0.360	1.000	0.596	1.000
Finland	119	0.842	0.246	1.000	0.800	1.000
France	604	0.930	0.189	1.000	1.000	1.000
Germany	690	0.842	0.267	1.000	0.709	1.000
Ireland	68	0.811	0.321	1.000	0.683	1.000
Italy	204	0.743	0.337	0.971	0.548	1.000
Norway	149	0.776	0.341	1.000	0.532	1.000
Portugal	86	0.924	0.218	1.000	1.000	1.000
Spain	610	0.941	0.178	1.000	1.000	1.000
Sweden	244	0.790	0.339	1.000	0.526	1.000
Switzerland	189	0.740	0.290	0.830	0.468	1.000
UK	1,628	0.888	0.228	1.000	0.907	1.000
Total	4,806	0.868	0.255	1.000	0.852	1.000

Table 9: cash-flow rights, control rights, and their ratio.

Table 8 (continued)

Country	Number of firms	<i>Own = 20% Con</i>	Dual class shares (%)	Pyramids	Holdings through multiple control chains	Controlling owner alone
<i>Panel C: Firms controlled by a widely held financial firm</i>						
Austria	9	18.29	55.56	22.22	22.22	88.89
Belgium	16	20.00	0.00	0.00	0.00	93.75
Finland	0	n.a.	n.a.	n.a.	n.a.	n.a.
France	69	19.94	2.90	15.94	1.45	60.87
Germany	65	18.85	21.54	53.85	40.00	33.85
Ireland	3	20.00	0.00	33.33	33.33	100.00
Italy	27	18.39	33.33	29.63	14.81	40.74
Norway	5	20.00	0.00	40.00	20.00	40.00
Portugal	3	20.00	0.00	0.00	0.00	0.00
Spain	74	20.00	0.00	20.27	9.46	51.35
Sweden	6	13.72	60.00	33.33	33.33	16.67
Switzerland	19	15.51	31.58	15.79	0.00	68.42
UK	166	18.77	31.93	27.71	18.67	35.54

Table 8, part 2: control-enhancing devices for widely held financial firm controllers.

Table 10

How concentrated is control by families/unlisted firms? This table provides summary statistics on the concentration of control by families and unlisted firms. *Average number of firms per family* refers only to firms in the sample. *Percent of total market value of listed corporate assets that families control* is the aggregate market value of common equity of firms controlled by the largest, the top 5, the top 10, and the top 15 families divided by the total market value of common equity for all firms included in the sample for a given country.

Country	Average number of firms per family	Percent of total market value of listed corporate assets that families control			
		Top family	Top 5 families	Top 10 families	Top 15 families
Austria	1.16	5.64	15.59	19.47	22.15
Belgium	1.22	5.63	20.38	30.45	36.63
Finland	1.05	3.94	13.98	21.82	25.90
France	1.18	5.94	22.04	29.18	33.80
Germany	1.24	5.43	15.66	21.29	25.01
Ireland	1.16	3.40	11.88	14.46	15.38
Italy	1.46	10.40	16.83	20.18	21.92
Norway	1.29	3.73	16.01	23.13	27.18
Portugal	1.23	6.11	24.59	34.23	36.77
Spain	1.19	1.66	6.97	10.92	13.48
Sweden	1.27	3.37	9.29	13.26	15.66
Switzerland	1.10	17.89	24.35	28.94	30.93

Table 10: concentration of family control by market value.

7 Economic interpretation

7.1 The main agency problem differs across governance systems

In a dispersed ownership system, the primary agency problem is often between managers and shareholders. In a controlling-owner system, the primary agency problem is often between controlling shareholders and minority shareholders. Faccio and Lang show that much of continental Europe belongs closer to the second system.

7.2 Family control can be stabilizing and problematic at the same time

A controlling family may provide long horizons, monitoring, and continuity. But when control exceeds cash-flow ownership, the family captures private benefits while bearing only part of the economic cost. The paper does not estimate private benefits directly, but it maps the control structures that make such benefits possible.

7.3 Country law matters, but law is not enough

The dual class share evidence shows that legal restrictions influence control separation. Yet country practice remains highly heterogeneous even conditional on legal category. Ownership structures are therefore shaped by law, market history, exchange rules, privatization histories, family business groups, and norms about control.

7.4 Large-firm samples can mislead

Because large firms are more likely to be widely held, a sample of only the 20 largest firms can make European markets appear more dispersed than they are. Including medium and small listed firms reveals much more family control. This is one of the paper's most important empirical lessons for corporate governance research design.

8 Limitations and caveats

8.1 Data are cross-sectional and late-1990s centered

The paper provides a detailed ownership snapshot, not a long panel of dynamic ownership changes. The ownership data are from 1996 to 1999 depending on the country. This is appropriate for measuring ultimate ownership structures, which tend to be stable, but it limits time-series analysis.

8.2 Unlisted firms are often treated as families

When the authors cannot trace the owners of an unlisted company, they classify it as family controlled. They provide empirical justification for this convention, but some misclassification is possible. The likely alternatives - state, widely held corporations, or widely held financial institutions - appear relatively uncommon among unlisted controllers in the countries for which they can check.

8.3 A 20% threshold is a convention

The main threshold is 20%, following earlier literature. This is a plausible threshold for effective control, but not a structural estimate of control power. The paper partially addresses this by discussing 10% thresholds, but actual control can depend on shareholder turnout, coalition formation, voting caps, and firm-specific governance rules.

8.4 The paper maps governance structures rather than outcomes

The paper documents who controls firms and how control is enhanced. It does not directly estimate effects on valuation, investment, payout, innovation, or minority expropriation. Its contribution is foundational measurement, which later papers can use to study consequences.

9 Takeaways

1. Western European listed corporations are not predominantly Berle-Means firms. Family control and widely held ownership coexist, with a strong Anglo-continental split.
2. Ultimate ownership differs from direct ownership; tracing control chains is essential.
3. Cash-flow rights and voting rights can differ materially, especially in Belgium, Italy, Norway, Sweden, and Switzerland.
4. Dual class shares, pyramids, and multiple control chains matter, but their importance varies across countries and owner types.
5. Family control often coincides with family management, making family governance a central feature of European corporate governance.
6. Large firms are more likely to be widely held, so conclusions based only on large firms understate the prevalence of family control.
7. The paper's incremental contribution is a broad, carefully traced ultimate-ownership database for Western Europe and a mechanism-level account of how control is separated from cash-flow ownership.